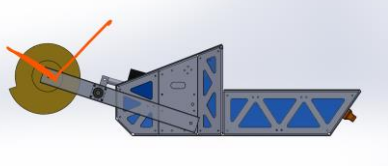
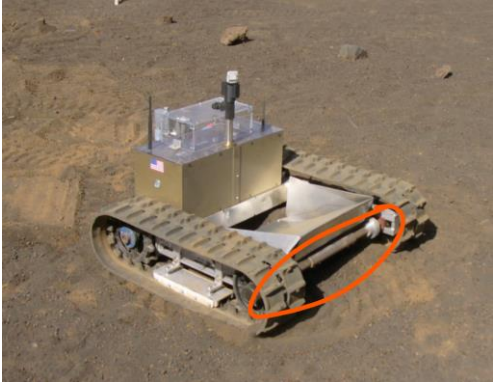
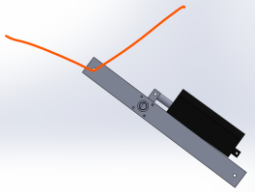


Different Scraper Design Ideas:

Jason's Preferences: Design 1, 7, 6 (in order of preference)

No.	Design	Pros	Cons
1	Using a similar system to the bucket drum, but with slight differences: - Use the linear actuation system to lower and raise the scraper - Use the motor drive arm to tilt/rotate the scraper to deposit the regolith  (NB: this design is a combination of Cratos x Nova ARC payload, cratos utilises a rotating shaft) 	- Easy to implement into the current ARC payload - Similar to the bucket drum system = hopefully less work required - Easy work around to the deposition issue - Won't need another payload just for this system	- Will the motor drive arm be strong enough to rotate the regolith ? - Will the motor drive arm be driven by the regolith ?
2	A system that utilises only an actuation system to collect and deposit regolith. The system will deposit regolith when it retracts fully up. This would mean that a different sci payload (maybe made out of wood) would be required, with a hole at the front for regolith to fall through 	Won't need a rotation mechanism to deposit regolith	- Need to also make a new blackbox science payload - May be difficult to design this science payload - May be difficult/impossible to find a good spot that can both collect and deposit regolith ? - Dust mitigation of new science

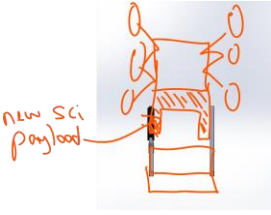
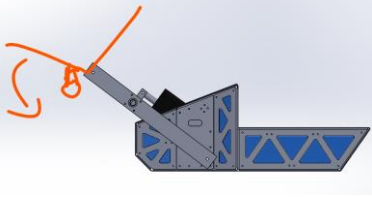
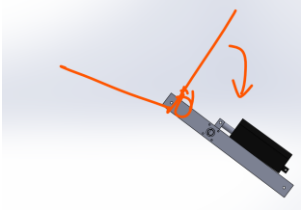
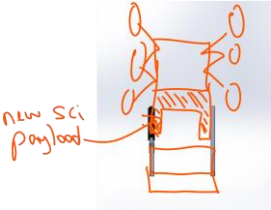
Commented [1]: the chain, sprocket, motor system used in the bucket drum to rotate the scraper to deposit regolith

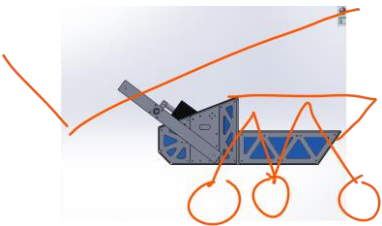
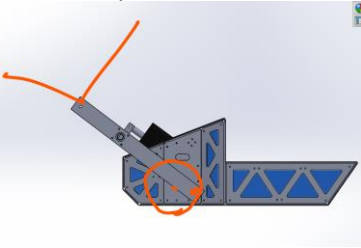
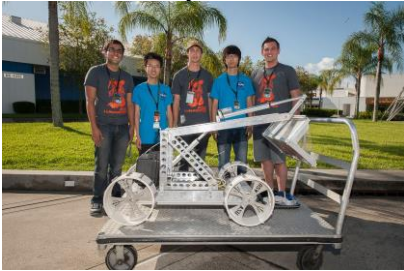
may need to change it tho so there is a shaft that is rotating instead

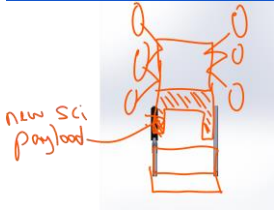
Commented [2]: Probs need to test wether those motors/chain system is strong enough to rotate the scraper - I am doubtful but who knows.

Commented [3]: true... I mean potentially could use a different/stronger motor for my system or potentially a gear reduction system could be implemented maybe to increase strength/force (again not entirely sure tho myself)

Commented [4]: probably fine tho cause if Cratos can do it

			payload may be an issue?
3	A system that uses an actuation system to collect regolith and a trap door on the bottom plate that will rotate and open to release the regolith. 	• Don't need a science payload	• Will the motor that holds the trap door be driven by the regolith as we collect ? • Most but not all regolith would be deposited (some would get caught on the trap door rotation system?)
4	A system that uses an actuation system to collect regolith and a trap door on the top plate that will rotate and open to release the regolith.  	• Better than design 2 because it'll be easier to find a spot/position for the linear actuation that can just collect the system	• Need to also make a new blackbox science payload • May be difficult to design this science payload • collect and deposit regolith ? • Dust mitigation of new science payload may be an issue? • Most but not all regolith would be deposited (some would get caught on the trap door rotation system?)
5	A system that uses an actuation system to collect and deposit regolith and a massive scrapper to deposit it over the science payload.	• Do not have to create a new science blackbox payload like Design 2 or 4.	• Not feasible to have such a long scraper lol • Not all regolith would be deposited

			
6	Use an actuation system to collect and deposit regolith. However, depositing of regolith will occur from tipping the scrapper to the front. The actuation arm will be attached to a wheel. Once regolith has been collected, this wheel will rotate the position of the regolith arm, allowing for regolith to be deposited. 	Avoids the issue of the motor potentially being driven by the regolith like designs 1 and 3	Might not have enough space on the side of the science box to have this rotation system (if so, a work around could be to make a new science payload, but then issues with this are presented above)
7	Double actuation system 	Could work well potentially - this team has done reasonably well in the Lunabotics	- Need to also make a new blackbox science payload - Dust mitigation of new science payload may be an issue?
8	Utilises a similar system to cratos, whereby it rotates around one shaft to collect regolith. To deposit regolith it rotates the other way and deposits it through the slot	- CRATTOSSS, what a beast - Low to the ground	- Sand may get clumped up under the sci payload - Not all regolith excavated will be deposited in this manner - Cannot utilise a ramp to deposit the regolith (unlike cratos which does), hence, sand would need to be deposited on



- the inside
- Need to also make a new blackbox science payload
 - Dust mitigation of new science payload may be an issue?